

Is Your AI Truly Yours? Leveraging Blockchain for Copyrights, Provenance, and Lineage

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Abstract—As Artificial Intelligence (AI) integrates into diverse areas, particularly in content generation, ensuring rightful ownership and ethical use becomes paramount. AI service providers are expected to prioritize responsibly sourcing training data and obtaining licenses from data owners. However, existing studies primarily center on safeguarding static copyrights, which simply treat metadata/datasets as non-fungible items with transferable/trading capabilities, neglecting the dynamic nature of training procedures that can shape an ongoing trajectory. In this paper, we present IBIS, a blockchain-based framework tailored for AI model training workflows. Our design can dynamically manage copyright compliance and data provenance in decentralized AI model training processes, ensuring that intellectual property rights are respected throughout iterative model enhancements and licensing updates. Technically, IBIS integrates on-chain registries for datasets, licenses and models, alongside off-chain signing services to facilitate collaboration among multiple participants. Further, IBIS provides APIs designed for seamless integration with existing contract management software, minimizing disruptions to established model training processes. We implement IBIS using Daml on the Canton blockchain. Evaluation results showcase the feasibility and scalability of IBIS across varying numbers of users, datasets, models, and licenses.

Index Terms—AI, blockchain, license, provenance, trust.

I. INTRODUCTION

ARTIFICIAL Intelligence (AI) technologies [1], [2] have increasingly permeated various aspects of daily life, spanning from information retrieval [3], [4] to content generation [5], [6]. Concurrently, AI service providers have made strides in commercializing their services. Nevertheless, as AI models rely on extensive datasets aggregated from diverse sources for training [7], [8], apprehensions have emerged regarding the potential infringement of copyrights [9], [10], [11] during the data acquirement and model training process. To uphold responsible and ethical AI practices [12], [13], comply with regulations, and reduce legal liabilities, AI service providers must actively collaborate with data owners, including content creators and media industry stakeholders. Establishing

licensing agreements [14], [15] and obtaining consent before utilizing data for AI model training is a key element of this collaboration [16]. Hence, there is a growing need for new frameworks addressing data provenance, lineage, and copyright compliance in the AI industry, tailored to its distinct needs and workflows.

However, addressing the concerns of AI data provenance and copyright compliance can be a nontrivial task, particularly when the entire training process occurs locally or within a black-box cloud service [17], limiting transparency for users. To bridge this gap, we harness the properties of blockchain technology, which offers a tamper-proof and trustworthy environment [18] to establish authenticity, provenance, and lineage [19], [20]. Owing to its inherent characteristics of immutability and transparency, blockchain has garnered widespread recognition as a suitable technology for achieving regulatory compliance [22], [23]. For instance, data recorded on the blockchain is digitally signed and inherently tamper-proof, thereby constituting an authentic and persistent record that accurately reflects an event(s) at a specific point in time. This makes blockchain a fitting candidate to address concerns related to data provenance and copyright compliance within the AI industry.

We have identified a series of functional challenges for such a blockchain-based compliance framework [27], [28], [29]: (i) The framework needs to be designed to seamlessly integrate with the existing workflow of AI model training [30], [31], [32]. (ii) The framework should support continuous model retraining and fine-tuning with new datasets [33], allowing for the generation of updated models while maintaining data provenance and lineage. (iii) The framework should support mechanisms for license expiration and renewal, accommodating diverse business models employed by data owners. (iv) The ownership of datasets and models [34], [35], along with all training actions, should be accompanied by evidence to clarify their licensing scope and ensure accountability for any subsequent actions. (v) The framework should facilitate communication between AI service providers and data owners [19], [36], enabling efficient attainment and documentation of licensing agreements. (vi) The framework should ensure the effective management and commercial sensitivity of licenses, safeguarding them against unauthorized access by third parties [37].

In this paper, we design, implement, and evaluate IBIS, a blockchain-based framework for data and model copyright management, provenance, and lineage in AI model training processes. IBIS empowers model owners to establish the provenance and lineage of their AI models and training datasets throughout retraining and fine-tuning processes, efficiently obtaining copyright licenses from the relevant copyright holders, and securely recording and renewing bilaterally signed copyright licenses as

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